Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

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| **Title of Proposed Technology** | *e.g. A technology that allows maintaining temperature fluctuations within a range of ±3℃.*<br>Inspection-Grade Digital Twin Technology for Aerospace and Marine Assets Using Rapid 3D Reconstruction | | | |
| **Korea's 12 National Strategic Technology Fields** | Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| **Industrial Technology Classification** [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | *e.g. Machinery and materials*<br>**Knowledge Services** | | **Information & Communication** | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | *e.g. Precision manufacturing machinery*<br>Human Resource Competency Development Services | *e.g. Cutting/processing machinery*<br>Human–System Interaction Technologies | Digital Content | Virtual Reality |
| **Concept of Proposed Technology** | *Describe the core concept, main functions, and potential uses of the proposed technology*<br>The core concept of the proposed technology is to enable **rapid, inspection-grade 3D digital twin generation for aerospace and marine assets** by converting real-world structures into high-fidelity 3D representations directly from scan data. The technology is designed to address the fundamental limitations of current aerospace and marine inspection practices, which still rely heavily on **2D** | | | |

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| | **drawings and fragmented inspection images** despite the increasing complexity and safety-critical nature of modern aircraft and marine systems.<br>In aerospace and marine environments, accurate understanding of **as-built conditions** is essential for non-destructive testing (NDT), Maintenance, Repair, and Overhaul (MRO), and certification-driven decision-making [1-3]. However, CAD-based digital twins often fail to reflect deviations from design, and conventional 3D reconstruction methods are too slow or computationally intensive for practical field deployment [4-7]. In addition, existing workflows lack intelligent mechanisms to compare reconstructed geometry with existing CAD or design references, making it difficult to systematically identify as-built deviations and assess inspection completeness. The proposed technology addresses this gap by leveraging **3D Gaussian Splatting**, an explicit 3D representation that enables fast reconstruction, real-time visualization, and robustness to incomplete or noisy scan data commonly encountered in hangars, shipyards, and offshore inspection environments.<br>The main functions of the proposed technology include:<br>• **Rapid generation of 3D digital twins** of aerospace and marine components and structures from multi-view image data, without reliance on pre-existing CAD models.<br>• **High-fidelity visualization of complex geometries**, including welded joints, curved surfaces, thin-walled structures, and densely assembled components typical of aircraft and marine systems.<br>• **Inspection-oriented 3D representation**, allowing inspectors to intuitively understand spatial relationships and access critical inspection regions that are difficult to interpret using 2D documentation alone.<br>• **Integration of non-destructive testing (NDT) data** within the 3D digital twin, enabling spatial localization and contextual visualization of inspection findings.<br>• **Interactive exploration and documentation**, supporting annotation, comparison across inspection cycles, and digital archiving required for certification and traceability.<br>• **AI-based comparison and alignment between reconstructed 3D digital twins and existing CAD or design data**, enabling detection and visualization of as-built deviations and inspection gaps.<br>Potential uses of the proposed technology include **aerospace and marine inspection and MRO operations**, such as inspection of aircraft structures, marine vessels, offshore platforms, and safety-critical welded assemblies. The technology also supports certification preparation, remote expert review, and long-term asset lifecycle management by providing a reliable and shareable 3D representation of actual as-inspected conditions. By enabling fast and accurate digital twins, the proposed technology improves inspection reliability, reduces dependence on expert intuition, and enhances safety assurance in aerospace and marine environments. |
| **Rationale for Joint R&D Project** | *Describe the need for this joint R&D project and why collaboration is required*<br>Aerospace and marine industries operate under **strict safety, reliability, and certification requirements**, where accurate understanding of structural conditions is essential for NDT, MRO, and regulatory compliance [8, 9]. Despite these requirements, inspection practices in both sectors still rely heavily on **2D engineering drawings and fragmented inspection images**, which are insufficient for representing complex geometries such as welded joints, curved surfaces, thin-walled structures, and densely assembled components [4-7]. This limitation increases dependence on expert judgment, complicates inspection traceability, and constrains effective lifecycle asset management [10].<br>Although digital twin technologies have been proposed as a solution, **existing approaches are not well suited to aerospace and marine field environments** [11]. CAD-based digital twins often fail to capture deviations between design and actual as-built conditions, while NeRF-based 3D reconstruction methods require long training times and high computational resources, making them impractical for time-sensitive inspection and MRO workflows [12, 13]. There is therefore a clear and urgent need for a **fast, robust, and inspection-oriented 3D digital twin technology** that can be generated directly from scan data and used immediately in hangars, shipyards, and offshore inspection settings. Furthermore, effective use of existing CAD data in inspection workflows requires intelligent processing capabilities. Direct manual comparison between 2D drawings, CAD models, |

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| | and real-world inspection images is inefficient and highly dependent on expert interpretation. **AI-based comparison and validation methods are therefore essential to automatically align reconstructed 3D digital twins with design references, identify discrepancies, and support systematic inspection and certification processes.** Developing such AI-driven inspection intelligence requires advanced research expertise in computer vision, geometric learning, and data-driven modeling, combined with industrial validation using real inspection data - further reinforcing the need for joint R&D collaboration.<br>This joint R&D project addresses this need by developing a **3D Gaussian Splatting–based digital twin platform** specifically tailored to aerospace and marine inspection scenarios. The project targets rapid reconstruction, real-time visualization, and robustness to incomplete or noisy scan data, enabling practical deployment in safety-critical environments where access is limited and inspection windows are short. However, achieving this objective requires the integration of **advanced 3D reconstruction research, computer vision (CV)–based inspection intelligence, and real-world industrial validation**, which cannot be effectively realized by a single organization alone.<br>The **University of Toronto (U of T)** serves as the sole academic research lead for this project, providing internationally recognized expertise in CV, 3D reconstruction, and data-driven modeling. The U of T is responsible for the development of 3D Gaussian Splatting–based reconstruction algorithms, inspection-oriented visualization methods, and quantitative validation frameworks required to meet aerospace and marine inspection standards. Its role ensures scientific rigor, algorithmic innovation, and reproducible performance evaluation that cannot be achieved through industry-led development alone.<br>The Korean SME provides essential industrial leadership and application context, including access to real aerospace and marine inspection use cases, non-destructive testing workflows, and system integration requirements necessary for commercialization and field deployment. The SME's involvement ensures that the developed technology directly addresses operational constraints, inspection time limitations, and certification-driven documentation requirements encountered in real-world aerospace and marine environments.<br>Collaboration between the U of T and the Korean SME is therefore essential to bridge the gap between **state-of-the-art academic research** and **practical aerospace and marine inspection needs**. This joint R&D structure enables the development of a deployable digital twin technology that improves inspection reliability, enhances certification traceability, and supports safer and more efficient lifecycle management of aerospace and marine assets. |
| **Key Words** | *Provide key terms that describe the proposed technology and its main features (min 3, max 5)*<br>1. Aerospace and Marine Digital Twin<br>2. 3D Gaussian Splatting–Based Digital Twin<br>3. Inspection-Oriented 3D Reconstruction<br>4. Non-Destructive Testing (NDT) Visualization<br>5. MRO and Certification Support Technology |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level ( 2 ) | End of the Project | Level ( 6) |
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| **R&D Objectives** | *Describe the specific technologies you aim to develop through this R&D project*<br>This joint R&D project aims to develop **inspection-grade digital twin technologies** for aerospace and marine environments, where safety-critical assets and strict certification requirements demand accurate representation of as-built conditions. The University of Toronto (U of T) leads core algorithm development, inspection intelligence, and validation, in collaboration with a Korean SME for field deployment and commercialization.<br>The project develops a **3D Gaussian Splatting–based rapid 3D reconstruction technology** that enables high-fidelity digital twins of aircraft, marine vessels, and offshore components directly from multi-view images, without reliance on CAD models. The approach is optimized for complex geometries (e.g., thin-walled structures, welded assemblies) and challenging field conditions, with emphasis on fast reconstruction and real-time rendering for time-constrained inspection and MRO workflows.<br>In parallel, the project advances **CV-based inspection intelligence**, including multi-view processing and inspection-oriented visual enhancement, to improve interpretation of critical structural elements beyond traditional 2D documentation. The digital twins further integrate **NDT data** through spatial registration and visualization, supporting defect localization, traceability, and certification-driven decision-making.<br>The project also delivers **interactive visualization and analysis tools** for real-time navigation, annotation, temporal comparison, and remote expert review in hangars, shipyards, and offshore environments, alongside **quantitative validation frameworks** aligned with aerospace and marine requirements.<br>In addition, **AI-based digital twin enhancement and CAD comparison technologies** enable automated alignment with design references, detection of design-to-as-built deviations, and assessment of reconstruction completeness, improving inspection efficiency, reducing manual interpretation, and strengthening certification traceability. | | | |
| **R&D Target Specifications** | *Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters*<br>The target performance specifications of this R&D project are defined to ensure that the proposed digital twin technology meets **inspection-grade, safety-critical requirements** in aerospace and marine environments. The specifications are structured to cover **geometric accuracy, computational performance, algorithmic robustness, data integration, and system-level reliability**, with all targets being quantitatively measurable and experimentally verifiable.<br>**1. Geometric Reconstruction Accuracy**<br>The project targets **inspection-grade geometric fidelity** suitable for aerospace and marine MRO applications. Using 3D Gaussian Splatting–based reconstruction, the digital twin will achieve:<br>• **Mean surface deviation ≤ 5 mm**<br>• **95th percentile surface error ≤ 10 mm**<br>evaluated against high-precision reference geometry. These targets support defect localization, weld inspection, and comparison across inspection cycles.<br>**2. Reconstruction Efficiency and Computational Performance**<br>To enable practical field deployment, the reconstruction pipeline targets:<br>• End-to-end reconstruction time **≤ 15 minutes per inspection target**<br>• Gaussian optimization iterations **≤ 30,000**<br>• GPU memory usage **≤ 12 GB (workstation-class hardware)**<br>while maintaining inspection-grade fidelity.<br>**3. Real-Time Rendering and Interactive Performance**<br>For effective inspection and analysis, the system supports: | | | |

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| | • Real-time rendering **≥ 30 FPS at 1080p resolution**<br>• Camera navigation latency **≤ 50 ms end-to-end**<br>• Dynamic level-of-detail (LOD) control for large or dense structures<br>This ensures that inspectors can interactively explore complex aircraft or marine structures without perceptual lag.<br>**4. Robustness to Field Data Limitations**<br>To accommodate realistic inspection constraints, the system targets:<br>• Tolerable missing view ratio: **Up to 30% occluded or unobserved surface areas**<br>• Minimum image set for stable reconstruction: **50–70 calibrated images per asset**<br>• Robust performance under variable lighting and reflective metallic surfaces.<br>**5. Inspection-Oriented Feature Resolution**<br>To preserve inspection-relevant details:<br>• Minimum resolvable feature size **≤ 10 mm for surface features (e.g., weld seams, joints)**<br>• Consistent visual contrast across Gaussian primitives to avoid inspection ambiguity<br>These specifications ensure that the digital twin supports visual inspection and spatial reasoning, even if it is not intended for micrometer-level metrology.<br>**6. NDT Data Integration and Spatial Registration Accuracy**<br>For inspection traceability and certification:<br>• 2D NDT-to-3D digital twin registration error **≤ 5 mm RMS**<br>• Repeatability of registration across inspection cycles **≤ 3 mm variance**<br>This enables reliable defect traceability, comparison over time, and certification documentation.<br>**7. System Stability, Repeatability, and Validation Metrics**<br>To ensure reliability in safety-critical use:<br>• Reconstruction repeatability (same asset, repeated scans) **≤ 5 mm mean deviation**<br>• System uptime **≥ 99% (excluding planned maintenance)**<br>• No critical system failures during continuous operation for **≥ 8 hours** of inspection activity<br>**8. Inspection Workflow Effectiveness (System-Level Metrics)**<br>System-level inspection outcomes target:<br>• **≥ 30% reduction** in inspection interpretation time<br>• **≥ 25% improvement** in defect localization accuracy<br>• **≥ 20% improvement** in inter-rater agreement<br>**9. AI-Based CAD Comparison and Discrepancy Detection Performance**<br>• CAD-to-digital twin alignment error **≤ 5 mm RMS**<br>• **≥ 90% true positive rate** for design–as-built deviation detection<br>• Automated discrepancy reporting latency **≤ 5 seconds** |
| **R&D Plan** | *Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives*<br>This R&D project adopts a **validation-driven, inspection-centered approach** to develop a rapid 3D digital twin platform based on 3D Gaussian Splatting for aerospace and marine inspection, MRO, and NDT environments. The project advances from **TRL 2 to TRL 5** by integrating scan-based 3D reconstruction, CV-based inspection intelligence, and inspection-oriented visualization with quantitative, reference-based validation. The U of T leads algorithm development and independent validation, while the Korean SME leads system integration, field deployment, and commercialization.<br>**Research Approach**<br>The approach prioritizes inspection usability over visual realism, enabling spatial understanding, defect localization, and certification traceability. Multi-view images are reconstructed into explicit 3D representations using 3D Gaussian Splatting for fast convergence and real-time visualization. CV modules enhance inspection-relevant features and integrate NDT data within the 3D frame. Performance is validated against reference geometry and baseline workflows using repeatability, robustness, and inspection task metrics.<br>**Phase 1: Requirements Definition and System Architecture Design (Months 1–4, TRL 2→3)**<br>**Key Tasks** |

- Define representative aerospace and marine inspection scenarios, including welded assemblies, curved thin-walled structures, confined-access regions, and reflective metallic surfaces.
- Specify quantitative performance targets for reconstruction accuracy, reconstruction time, rendering performance, robustness to occlusion, and NDT-to-3D alignment.
- Define inspection-driven quality criteria, including defect localization precision, annotation traceability, and comparison across inspection cycles.
- Design the end-to-end system architecture, specifying data acquisition, calibration, 3D reconstruction, inspection visualization, and validation pipelines.

**Methodologies**

- Inspection workflow analysis and human-centered task modeling for aerospace and marine MRO.
- System engineering and computational budget analysis for field-deployable reconstruction and visualization.
- Definition of geometry-level (surface error), system-level (time, FPS), and task-level (inspection accuracy, time) validation criteria.

**Expected Milestones**

- Aerospace/marine inspection requirements and quantitative performance targets finalized.
- System architecture and validation framework approved and frozen for Phase 2 implementation.

**Phase 2: 3D Gaussian Splatting Core Development and Pilot Validation (Months 5–10, TRL 3)**

**Key Tasks**

- Develop the 3D Gaussian Splatting reconstruction core for aerospace and marine assets using calibrated multi-view images.
- Implement adaptive Gaussian initialization, densification, pruning, and regularization strategies for thin-walled structures, curved surfaces, and long weld seams.
- Address reflective metallic materials and partial occlusions common in hangar and shipyard environments.
- Conduct pilot reconstruction validation using representative aerospace and marine components.

**Methodologies**

- Algorithm development with controlled benchmarking against reference geometry.
- Quantitative evaluation using mean/95th percentile surface error, reconstruction time, GPU memory usage, and rendering FPS.
- Failure-mode analysis covering glare, sparse overlap, occlusion, and low-texture regions.

**Expected Milestones**

- Reconstruction engine v1 validated on representative assets.
- Demonstrated fast reconstruction (≤15 minutes per target) with stable interactive rendering (≥30 FPS).

**Phase 3: AI-enabled inspection intelligence and NDT integration (Months 11–18, TRL 3 → 4)**

**Key Tasks**

- Develop CV modules to enhance inspection-relevant features (weld seams, joints, fasteners) in reconstructed 3D twins.
- Implement inspection-oriented visualization and annotation tools aligned with aerospace and marine certification documentation.
- Develop spatial registration and visualization of 2D NDT data (e.g., ultrasonic or radiographic inspection results) within the 3D digital twin.
- Evaluate inspection performance improvements relative to 2D drawing–based workflows.
- Develop AI-based methods for aligning reconstructed 3D digital twins with existing CAD or design references and automatically identifying as-built deviations.

**Methodologies**

- Development of multi-view feature enhancement, capture quality assessment, and inspection annotation tools led by the U of T.
- Controlled inspection studies with trained users to measure interpretation time, defect localization accuracy, and inter-rater consistency.
- Statistical evaluation of task-level performance and robustness across inspection scenarios.
- Development of AI-based digital twin enhancement and CAD comparison methods, including

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| | learning-based alignment between reconstructed 3D Gaussian Splatting representations and available CAD or design reference models, to assess reconstruction completeness and identify design-to-as-built deviations.<br>**Expected Milestones**<br>• Integrated inspection-intelligent digital twin prototype v1 achieving TRL 4.<br>• Demonstrated ≥30% reduction in inspection interpretation time and ≥25% improvement in defect localization accuracy compared to baseline methods.<br>• AI-based CAD alignment and comparison module v1 capable of registering reconstructed digital twins with available CAD references and quantifying geometric deviations.<br>• Demonstration of automated identification of missing or low-confidence regions in reconstructed digital twins to support improved scan completeness and inspection reliability.<br>**Phase 4: System Hardening and Representative Validation (Months 19–24, TRL 4 → 5)**<br>**Key Tasks**<br>• Optimize system robustness, usability, and deployment readiness for aerospace and marine field environments.<br>• Conduct regression validation to ensure optimization does not degrade reconstruction accuracy or inspection performance.<br>• Perform representative environment trials in hangar and shipyard settings.<br>• Assess usability, inspector acceptance, and suitability for certification-oriented documentation.<br>**Methodologies**<br>• Extended-duration and repeated inspection trials under realistic access, lighting, and operational constraints.<br>• Comparative evaluation against existing inspection practices using time, accuracy, repeatability, and traceability metrics.<br>• Joint field validation led by the Korean SME with technical support from the U of T.<br>**Expected Milestones**<br>• Prototype v2 validated under representative aerospace and marine conditions.<br>• TRL 5 demonstration completed with comprehensive technical, validation, and deployment documentation.<br>**Overall Timeline Summary**<br>• Months 1–4: Requirements and system architecture definition (TRL 2 → 3)<br>• Months 5–10: 3D Gaussian Splatting core development and pilot validation (TRL 3)<br>• Months 11–18: Inspection intelligence and NDT integration with validation (TRL 3 → 4)<br>• Months 19–24: System hardening and representative environment validation (TRL 4 → 5) |

## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | *Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.*<br>The U of T serves as the academic research institute and principal research lead for this joint R&D project, responsible for core algorithm development, inspection-oriented system design, AI-based digital twin enhancement, and independent performance validation for aerospace and marine inspection applications. From a technical perspective, the U of T leads the development of key digital twin technologies, including 3D Gaussian Splatting–based reconstruction, CV-based inspection intelligence, AI-driven CAD comparison and discrepancy analysis, and inspection-oriented visualization. This work addresses complex aerospace and marine geometries (e.g., thin-walled structures, curved surfaces, welded assemblies) and real-world challenges such as reflective materials, occlusions, and sparse viewpoints.<br>The research scope covers the full development lifecycle, including inspection-driven requirements analysis, algorithm design and optimization, AI-based validation and quality assessment, system-level integration, and the establishment of quantitative evaluation frameworks aligned with aerospace and marine certification needs. The U of T defines inspection-grade performance metrics, validation |

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| | criteria, and benchmarking protocols to assess reconstruction fidelity, robustness under field conditions, inspection usability, and AI-supported deviation detection relative to conventional 2D-based practices.<br>The U of T provides specialized research personnel, high-performance computing resources, and laboratory infrastructure to support algorithm development, AI model training, and performance benchmarking. Specific responsibilities include:<br>Designing and implementing the 3D Gaussian Splatting reconstruction engine for aerospace and marine inspection scenarios.<br>• Developing CV-based inspection intelligence and inspection-oriented visualization.<br>• Establishing quantitative performance metrics and validation protocols aligned with certification requirements.<br>• Conducting controlled validation and comparative evaluation against baseline inspection methods.<br>• Leading algorithm- and system-level integration to ensure consistency, reliability, and scalability.<br>• Providing technical guidance and knowledge transfer to the Korean SME for system integration, field deployment, and commercialization readiness. |
| **Expected R&R of Korean SME Partner** | *Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.*<br>The Korean SME serves as the industrial partner and commercialization-oriented lead in this joint R&D project, providing application context, system integration capability, and field validation resources to translate advanced digital twin research into deployable aerospace and marine inspection solutions. The SME ensures that the developed technologies address real-world operational constraints, certification requirements, and market needs.<br>From a technical perspective, the Korean SME leads system-level engineering and integration to operationalize the inspection-grade digital twin platform. This includes defining deployment architectures, interfacing reconstruction and visualization software with inspection hardware and data acquisition systems, and ensuring compatibility with existing aerospace and marine inspection workflows. The SME contributes domain expertise in NDT, inspection reporting, and certification documentation to align research outputs with industry standards.<br>Key responsibilities include:<br>• Designing field-ready data acquisition workflows (camera setup, calibration, and capture protocols) for hangars, shipyards, and offshore environments.<br>• Integrating the U of T's reconstruction and inspection algorithms with practical user interfaces and deployment platforms.<br>• Supporting software engineering for system stability, usability, and deployment readiness, including workflow automation.<br>• Preparing the platform for scalable deployment, considering maintainability, portability, and interoperability.<br>The Korean SME leads representative environment testing and field-level validation by providing access to real inspection use cases, components, or facilities and supporting pilot trials under realistic operational conditions. In collaboration with the U of T, the SME evaluates system robustness, usability, and inspection effectiveness relative to conventional 2D-based practices. |

## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | *Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.*<br>This joint R&D project is expected to deliver significant technological and industrial impact by establishing an **inspection-grade, AI-enhanced digital twin platform** optimized for aerospace and marine environments. Led by the University of Toronto (U of T) with industrial deployment by a Korean SME, the project bridges advanced 3D reconstruction research and safety-critical inspection needs.<br>Technologically, the project delivers a **3D Gaussian Splatting–based digital twin** capable of rapid |

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| | reconstruction, real-time visualization, and inspection-oriented representation of aerospace and marine assets. **AI-based digital twin enhancement and CAD comparison** enable automated alignment with design references, detection of as-built deviations, and objective assessment of reconstruction completeness. The platform achieves inspection-grade accuracy without reliance on pre-existing CAD models, robust performance under realistic field conditions, and seamless integration of NDT data, representing a clear advancement over CAD-based and NeRF-based approaches. A TRL 5 demonstration confirms technical maturity and deployability.<br>From a commercialization perspective, the project generates a **field-ready, AI-supported inspection solution** for aerospace and marine markets, supporting deployment for MRO, integration with NDT systems, and adoption for lifecycle management and certification preparation. AI-driven comparison and quality assessment reduce manual interpretation, improve inspection confidence, and strengthen traceability, enabling early market adoption and scalability.<br>More broadly, the project enhances inspection reliability, reduces dependence on expert judgment, and improves safety assurance for complex, safety-critical assets. By enabling systematic, repeatable inspection decisions and efficient lifecycle asset management, the technology supports reduced downtime, lower maintenance costs, and increased operational efficiency. At a national and international level, the project strengthens competitiveness in AI-enabled digital inspection technologies and promotes academia–industry collaboration, workforce development, and long-term innovation in aerospace and marine sectors.<br>reliability standards in aerospace and marine sectors. |

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## ☐ Terms:

**1) Industrial Technology Classification**

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

**2) Technology Readiness Level (TRL)**

| Description | | |
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| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |